



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2020 - 2021)

JEE(Advanced)
REVIEW TEST
26-07-2020

JEE(Main + Advanced) : NURTURE COURSE (PHASE : TEPS)

PAPER-2

Time : 3 Hours

Maximum Marks : 186

READ THE INSTRUCTIONS CAREFULLY

GENERAL :

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **28** pages and that all the **18** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble  **COMPLETELY.**
12. The correct way of darkening a bubble is as : 
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take **$g = 10 \text{ m/s}^2$** unless otherwise stated.

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

Please see the last page of this booklet for rest of the instructions

SOME USEFUL CONSTANTS

Atomic No. : H = 1, B = 5, C = 6, N = 7, O = 8, F = 9, Al = 13, P = 15, S = 16,
Cl = 17, Br = 35, Xe = 54, Ce = 58

Atomic masses : H = 1, Li = 7, B = 11, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24,
Al = 27, P = 31, S = 32, Cl = 35.5, Ca = 40, Fe = 56, Br = 80, I = 127,
Xe = 131, Ba = 137, Ce = 140, Cu = 63.5, Ne = 20, K = 39, Mn = 55

Space for Rough Work

HAVE CONTROL → HAVE PATIENCE → HAVE CONFIDENCE ⇒ 100% SUCCESS

BEWARE OF NEGATIVE MARKING

PART-1 : PHYSICS

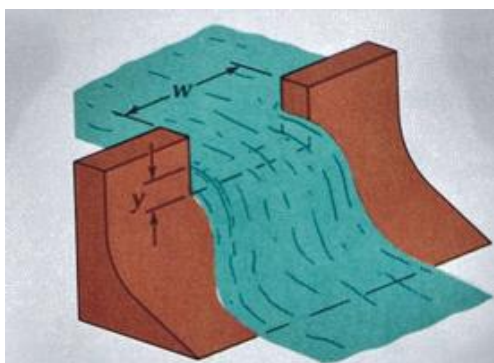
SECTION-I(i) : (Maximum Marks : 18)

- This section contains **SIX** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct.
- For each question, darken the bubble corresponding to the correct option in the ORS.
- For each question, marks will be awarded in one of the following categories :
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Zero Marks : 0 If none of the bubbles is darkened.
Negative Marks : -1 In all other cases

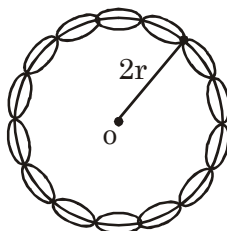
1. Ram decided to weigh a block he found near the Ganges river. For this he used symmetrical equal-arm scales, made of iron. In the air, weighing gave result $M = 2.1$ kg. When the scales were completely immersed in the water of the river, the result was different - for balancing iron weights of mass 1.8 kg in air needed to be put on block. In this case, the substance being weighed, and the weights were also completely immersed in water. What is the density of the block? Iron Density = 8 g/cm^3 , water density 1.0 g/cm^3 .
(A) $6/5 \text{ gm/cc}$. (B) $8/7 \text{ gm/cc}$. (C) $7/9 \text{ gm/cc}$. (D) None of these
2. With one mole of monatomic gas we conducted a four-step process thermodynamic cycle. During process 1-2, the gas does work of 24 J, while the temperature was directly proportional to volume to the power of n . Temperatures T_1 and T_2 are known, moreover $T_1 < T_2$. During 2-3 the gas expanded isothermally, in process 3-4 it expanded adiabatically, and in section 4-1 it was isothermally compressed. What was the work done by the gas during process 3-4?
(A) $24n \text{ J}$ (B) $12n \text{ J}$ (C) 24 J (D) $36n \text{ J}$

Space for Rough Work

3. This figure shows the diagram of a spillway from which water is flowing. The width of the spillway is w and height of water is y . What is the rate of volume flow through the spillway? Take $w = 6\text{m}$ and $y = 5\text{m}$.



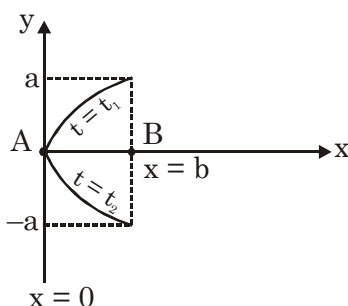
- (A) $200 \text{ m}^3/\text{sec}$ (B) $100 \text{ m}^3/\text{sec}$ (C) $\frac{200}{3} \text{ m}^3 / \text{sec}$ (D) $\frac{100}{3} \text{ m}^3 / \text{sec}$
4. A particle of mass m revolves in a circle of radius $2r$ on a frictionless horizontal surface. The necessary centripetal force for the motion is provided by a rubber band of natural length r and force constant k . Now the block is slightly perturbed by providing a small radial impulse. Calculate the time period of radial oscillations of the block.



- (A) $2\pi\sqrt{\frac{2m}{3k}}$ (B) $2\pi\sqrt{\frac{m}{2k}}$ (C) $2\pi\sqrt{\frac{2m}{k}}$ (D) $\pi\sqrt{\frac{m}{k}}$

Space for Rough Work

5. A sphere of brass released in a long liquid column attains a terminal speed v_0 . If the terminal speed attained by the sphere of marble of the same radius and released in the same liquid is nv_0 , then the value of n will be. Given: The specific gravities of brass, marble and the liquid are 8.5, 2.5 and 0.8 respectively.
- (A) 5/17 (B) 17/77 (C) 11/31 (D) 17/5
6. A part of a stationary wave between consecutive node A and antinode B has the graphic profile as shown in the diagram at two instants of time $t = t_1$ and $t = t_2$ where $(t_1 - t_2) = T/2$; where T is the periodic time of the wave. The stationary wave is formed by superposition of two identical waves. Then the amplitude A and wavelength λ of each wave are given by



- (A) $A = 2a, \lambda = 2b$ (B) $A = a/2, \lambda = 2b$ (C) $A = a/2, \lambda = 4b$ (D) $A = 2a, \lambda = 4b$

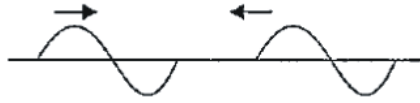
Space for Rough Work

SECTION-I(ii) : (Maximum Marks: 32)

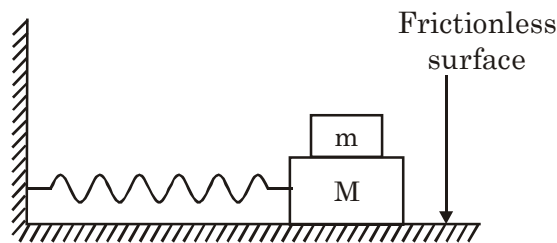
- This section contains **EIGHT** questions.
 - Each question has **FOUR** options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
 - For each question, choose the correct option(s) to answer the question.
 - Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.
Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct options.
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks : -2 In all other cases.
 - **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.
-
7. One mole of monoatomic gas is heated such that its molar heat capacity remains $2R$ during the process. If the volume of the gas doubles during the process,
- (A) Its temperature becomes one fourth.
(B) Its pressure doubles.
(C) work done by the gas is 0.5 times heat given.
(D) Change in internal energy is $3/4$ times the heat given.
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Space for Rough Work

8. Two identical sinusoidal pulses move in opposite directions on a stretched string. Kinetic energy of each pulse is k . Choose the **CORRECT** statement(s)



- (A) The total mechanical energy of each pulse at the moment shown is $2k$.
 (B) At the instant they overlap completely kinetic energy of the resulting pulse is $2k$.
 (C) At the instant they overlap completely the potential energy of resulting pulse is $4k$.
 (D) The mechanical energy at the moment shown is uniformly distributed along the pulse.
9. A block of mass M is connected to a spring of constant k . Its top surface is rough. A block of mass m is placed on it. The amplitude of oscillation A is such that the mass m is on the verge of slipping. Which of the following changes would still ensure that the mass m would oscillate with the block of mass M (keeping all the other parameters same)?



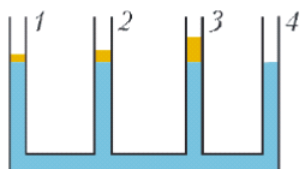
- (A) Increase k (B) Increase m (C) Increase M (D) Increase A

Space for Rough Work

10. Badminton shuttlecock has a mass of $m = 5 \text{ g}$ and cross-sectional area $S = 30 \text{ cm}^2$. When moving in air with speed v , the shuttle experiences a force resistance directed against velocity and equal to $F = \frac{5}{3} \times 10^{-2} v^2 r S$, where $r = 1 \text{ kg/m}^3$ is the air density. After being hit by a racket, the shuttlecock acquired a vertical speed of 20 m/s and reached a maximum height H .

Take $\ln \frac{7}{5} = 0.3$. (Here η = coefficient of viscosity)

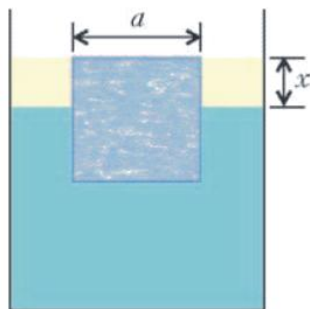
- (A) The coefficient $\frac{5}{3}$ has dimension of η .
 (B) The acceleration of the shuttlecock immediately after the strike is 14 m/s^2
 (C) The shuttlecock would fly up to a height of 20 m if its initial vertical speed was 10 m/s .
 (D) $H = 15 \text{ m}$.
11. There are four identical cylindrical vessels into which we have poured some water. We gently pour a layer of oil of thickness h , $2h$ and $3h$ respectively in 1st, 2nd, and 3rd branch. By what distance will the total fluid level in each vessel change compared to the initial position after equilibrium? Liquid does not spill out. Oil density is ρ_0 , water density ρ_1 ($\rho_1 > \rho_0$).



- (A) $\Delta h_1 = h \left(1 - \frac{\rho_0}{2\rho_1} \right)$ (B) $\Delta h_2 = 2h \left(1 - \frac{\rho_0}{2\rho_1} \right)$
- (C) $\Delta h_3 = 3h \left(1 - \frac{\rho_0}{2\rho_1} \right)$ (D) $\Delta h_4 = \frac{3}{2} h \frac{\rho_0}{\rho_1}$

Space for Rough Work

12. Water is poured into a cylindrical glass with a base area of $S = 400 \text{ cm}^2$; in which we immersed an ice cube with side $a = 2 \text{ cm}$ (fig.). Into the glass we pour kerosene of volume V_0 so that the upper surface of the cube coincides with kerosene surface. The cube is taken out of the glass, and another ice cube with side $b = 6 \text{ cm}$ is immersed in it. Some more volume V of kerosene must be poured into the glass, so that the cube is again covered to the top by kerosene? Consider that both cubes do not touch the bottom, and the liquid from the glass does not spills out. The densities of ice, water and kerosene are 0.9 g/cm^3 , 1.0 g/cm^3 and 0.8 g/cm^3 , respectively. Choose the **CORRECT** statement(s) :



(A) $V = 680 \text{ cc}$

(B) $V = 696 \text{ cc}$

(C) $V_0 = 396 \text{ cc}$

(D) $V_0 = 184 \text{ cc}$

Space for Rough Work

13. A wave pulse in a string is photographed at $t = 0$. The shape of the string is given by $y = A e^{-x^2}$. The string is known to have a tension T and mass per unit length of μ . The wave travels in positive x direction. Take $\int_0^\infty x^2 e^{-x^2} dx = \sqrt{\pi/4}$, $\sqrt{T/\mu} = 4$ m/s.
- (A) The pulse is symmetric about y axis at $t = 0$.
 (B) At $t = 0$ sec, the velocity of element at $x = \ln 2$ is downwards
 (C) The total energy of the pulse is $4\sqrt{\pi}TA^2$.
 (D) The rate at which energy is passing through point $x = 4$ m at $t = 1$ sec is zero.
14. A wire of density ρ is tapered so that its cross-sectional area varies with distance x (in m) according to the equation $A = (1.0 \times 10^{-3}x + 0.10)$ cm². A wave is sent along the wire in the positive x direction. The density of the wire is 3 gm/cc. Then if the wire is subject to a tension of 24.0 N,
- (A) The speed of the wave decreases as it propagates along the string.
 (B) The speed of the wave at the origin is $20\sqrt{2}$ m/s.
 (C) The speed of the wave at $x = 100$ m is 20 m/s.
 (D) The time taken by the wave to reach $x = 100$ m from $x = 0$ m is $\frac{5}{3}(2 - \sqrt{2})$

Space for Rough Work

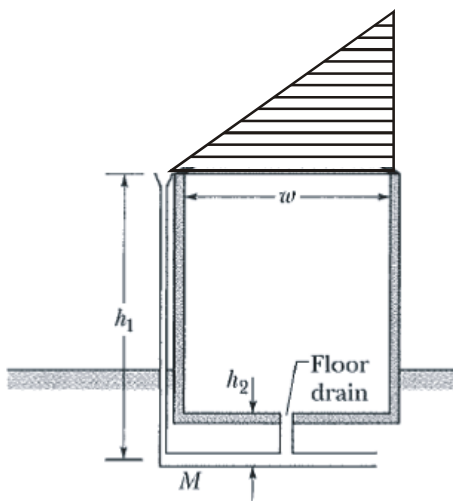
SECTION-I(iii) : (Maximum Marks : 12)

- This section contains **TWO** paragraphs.
- Based on each paragraph, there are **TWO** questions.
- Each question has **FOUR** options (A), (B), (C) and (D) **ONLY ONE** of these four options is correct.
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Paragraph for Questions 15 and 16

A very simplified schematic diagram of the rain drainage system for a home is shown in figure. Rain falling on the slanted roof runs off into gutters around the roof edge; it then drains through downspout into a main drainage pipe M below the basement, which carries the water to an even larger pipe below the street. In figure a floor drain in the basement is also connected to drainage pipe M. Suppose the following apply:

1. The downspout has height $h_1 = 11$ m,
2. The floor drain has height $h_2 = 1.0$ m,
3. Pipe M has area 10 cm^2 ,
4. The house has side width $w = 30$ m and front length $L = 60$ m,
5. All the water striking the roof goes through pipe M,
6. The wind speed is negligible (the rain falls vertically).

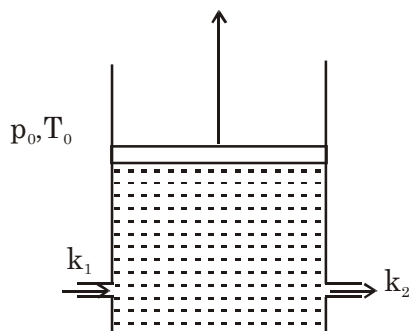


15. If the area of downspout is 10 cm^2 , and rain is falling at the rate of 1 cm/hr , what is the average rate of water flow through the downspout ?
 (A) 10 m/s (B) 5 m/s (C) 20 m/s (D) 8 m/s
16. At what minimum rainfall rate, in centimeters per hour, will water from pipe M reach the height of the floor drain and threaten to flood the basement?
 (A) 2.82 (B) 1.41 (C) 0.52 (D) 2.06

Space for Rough Work

Paragraph for Questions 17 and 18

There is 1 mole air in the cylinder under the massless piston at pressure p_0 . The cylinder walls have two valves: inlet K_1 and exhaust K_2 (figure). The inlet valve opens when the difference in air pressure outside and inside the cylinder exceeds $0.2p_0$, where p_0 is the atmospheric pressure. The outlet valve opens when the pressure difference inside and outside exceeds $0.4p_0$. The valves open for a short time till the pressure equalises & closes again. The piston oscillates very slowly so that the volume of air in the cylinder varies from V_0 to $2V_0$. The temperature outside and inside the cylinder is constant and equal to T_0 .



17. The valve k_1 opens n times while going up & valve k_2 opens m times while coming down.
 (A) $n = 4, m = 3$ (B) $n = 3, m = 2$ (C) $n = 2, m = 3$ (D) $n = 3, m = 1$
18. The maximum number of moles in cylinder is n_1 and minimum number of moles is n_2 choose the values.
 (A) $n_1 = \frac{216}{125}$ moles, $n_2 = 1$ moles. (B) $n_1 = \frac{125}{64}$ moles, $n_2 = \frac{3125}{3136}$ moles
 (C) $n_1 = \frac{125}{64}$ moles, $n_2 = 1$ moles (D) $n_1 = \frac{36}{25}$ moles, $n_2 = \frac{4}{5}$ moles

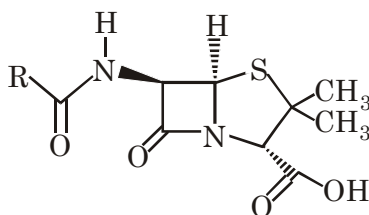
Space for Rough Work

PART-2 : CHEMISTRY

SECTION-I(i) : (Maximum Marks : 18)

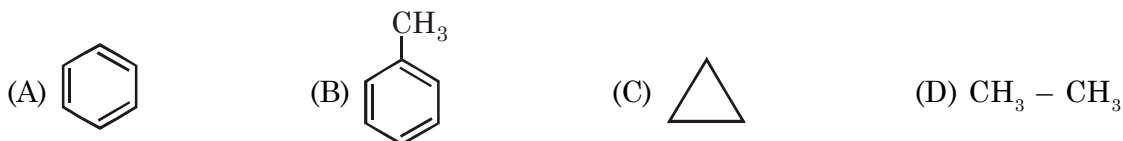
- This section contains **SIX** questions.
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1. How many chiral atom present in penicillin

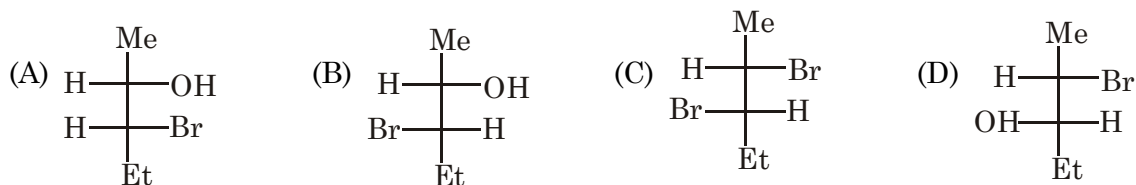


- (A) 4 (B) 5 (C) 6 (D) 3

2. Which of following decolourised Br_2 water solution :



3. Correct major product of reaction between cis pent-2-ene and Br_2 water is?

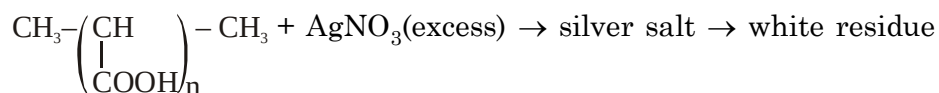


4. When 37 gm gaseous mixture of C_2H_6 and C_3H_8 gases is burnt completely in excess of O_2 gas, 110 gm CO_2 and 63 gm H_2O vapours are formed. The **INCORRECT** information related with the initial mixture is /are

- (A) Equal moles of both the gases were taken
 (B) Equal masses of both the gases were taken
 (C) Mass ratio of 'C' and 'H' atoms is 30 : 7
 (D) The total mass of oxygen used in combustion was 136 gm

Space for Rough Work

5. Consider the following set of reactions :



If 0.1 moles of silver salt is taken and weight of residue obtained is 54 gm, then what is the molecular mass of $\text{CH}_3 - \left(\begin{array}{c} \text{CH} \\ | \\ \text{COOH} \end{array} \right)_n - \text{CH}_3$? (Ag = 108)

- (A) $1030 \frac{\text{g}}{\text{mol}}$ (B) $495 \frac{\text{g}}{\text{mol}}$ (C) $320 \frac{\text{g}}{\text{mol}}$ (D) $108 \frac{\text{g}}{\text{mol}}$

6. Two flasks of equal volume are connected by a narrow tube (of negligible volume) all at 27°C and contain 0.7 mol of H_2 at 0.5 atm. One of the flasks (1) is then immersed into a bath kept at 127°C , while the other flask (2) remains at 27°C . The number of moles of H_2 in flask 1 and flask 2 are -

- (A) moles in flask 1 = 0.4 ; moles in flask 2 = 0.3
 (B) moles in flask 1 = 0.2 ; moles in flask 2 = 0.5
 (C) moles in flask 1 = 0.3 ; moles in flask 2 = 0.4
 (D) moles in flask 1 = 0.5 ; moles in flask 2 = 0.2

Space for Rough Work

SECTION-I(ii) : (Maximum Marks: 32)

- This section contains **EIGHT** questions.
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7. Select the **INCORRECT** option(s).

(A) Collision frequency (Z_{11}) becomes $\frac{1}{4}$ times of initial value, if volume is doubled at constant temperature

(B) Fraction of molecule in the range U_{mps} to $U_{\text{mps}} + 0.5\% U_{\text{mps}}$ is lesser for CH_4 than O_2 at same temperature.

(C) Molar volume $V_m > 22.4 \text{ L}$ at 1 atm and 273 K when attractive force dominate.

(D) The real gas behave as an ideal gas at critical point.

8. If a definite volume of '20 vol' H_2O_2 solution is diluted such that the volume of diluted solution becomes double than that of the original volume, then -

(A) The volume strength of diluted solution becomes '40 vol'

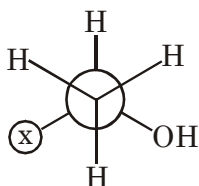
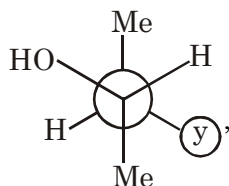
(B) The molarity of solution becomes half of its initial molarity

(C) The molality of solution becomes half of its initial molality

(D) The maximum amount of O_2 obtained from the solution remains the same.

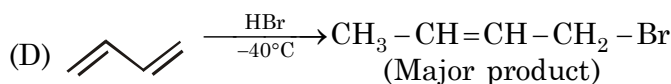
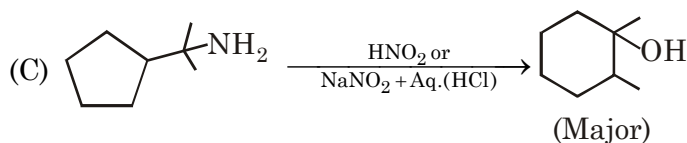
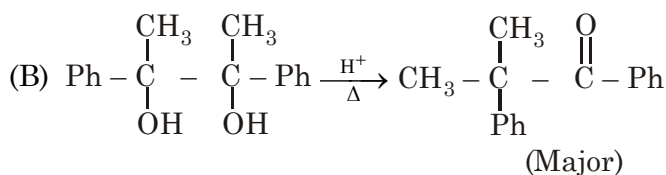
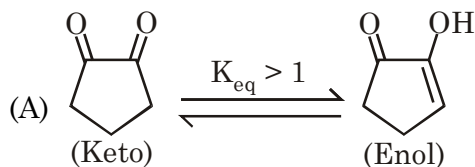
Space for Rough Work

9. Identify the incorrect statement(s) about the number of collisions of Ne atoms with the walls of container per unit time -
- (A) Increases when the temperature is increased at constant pressure
 (B) Remains the same when CO_2 is added to the container at constant temperature
 (C) Increases when CO_2 is added to the container at constant temperature
 (D) Decreases, when the average kinetic energy per molecule is decreased at constant volume.
10. For 2,3 butandiol possible value of x and y is/are :



- (A) H, (B) $\text{CH}_3 - \text{CH} - \text{OH}$ (C) OH (D) Me

11. Which of following option is/are correct?



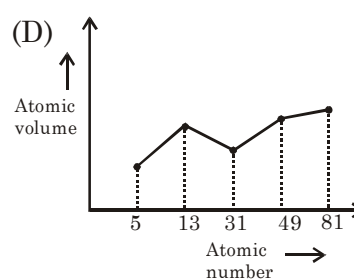
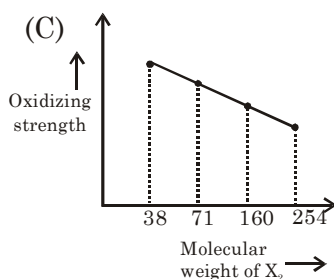
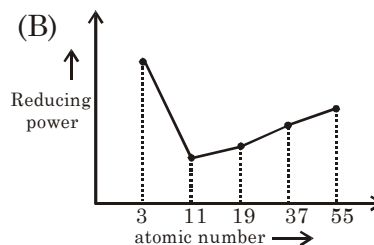
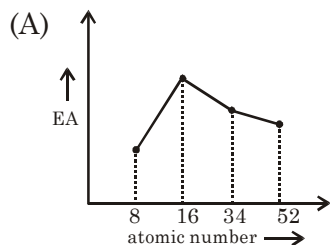
12. Which of following statement(s) is/are not true in aqueous ethanol?
- (A) $\text{CH}_3 - \text{CH}_2 - \text{S}^-$ is stronger base and stronger nucleophile than $\text{CH}_3 - \text{CH}_2 - \text{O}^-$.
 (B) $\text{CH}_3 - \text{CH}_2 - \text{S}^-$ is stronger base but less nucleophilic than $\text{CH}_3 - \text{CH}_2 - \text{O}^-$.
 (C) $\text{CH}_3 - \text{CH}_2 - \text{S}^-$ is a weaker base but more nucleophilic than $\text{CH}_3 - \text{CH}_2 - \text{O}^-$.
 (D) $\text{CH}_3 - \text{CH}_2 - \text{S}^-$ is both a weaker base and less nucleophilic than $\text{CH}_3 - \text{CH}_2 - \text{O}^-$.

Space for Rough Work

13. Which of the following statement(s) is/are correct?

- (A) Hydrogen bond is stronger than single H–H bond
- (B) Chloral hydrate shows intramolecular H-bonding
- (C) Dimer of CH_3COOH in benzene comprises of a 6 membered ring
- (D) C_3O_2 is planar & polar

14. Which of the following graph(s) against the indicated property(ies) is/are **CORRECT**?



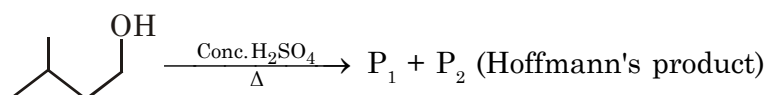
Space for Rough Work

SECTION-I(iii) : (Maximum Marks : 12)

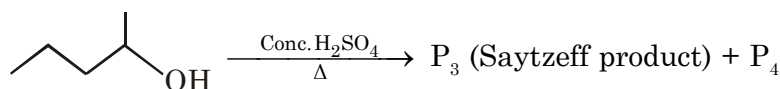
- This section contains **TWO** paragraphs.
- Based on each paragraph, there are **TWO** questions.
- Each question has **FOUR** options (A), (B), (C) and (D) **ONLY ONE** of these four options is correct.
- For each question, darken the bubble corresponding to the correct option in the ORS.
- For each question, marks will be awarded in one of the following categories :
Full Marks : +3 If only the bubble corresponding to the correct answer is darkened.
Zero Marks : 0 In all other cases.

Paragraph for Questions 15 and 16

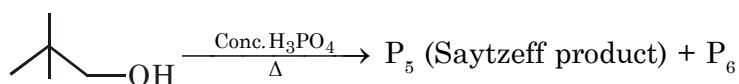
Reaction : 1



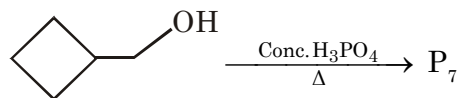
Reaction : 2



Reaction : 3



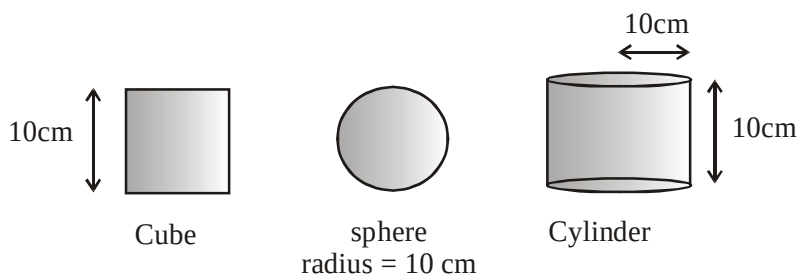
Reaction : 4



15. Which of following can show geometrical isomerism?
 (A) P_1 (B) P_2 (C) P_3 (D) P_4
16. Correct match is?
 (A) P_7 and P_6 : Ring chain isomer (B) P_1 and P_4 : Position isomer
 (C) P_3 and P_5 : Identical (D) None of these

Space for Rough Work

Paragraph for Questions 17 and 18



Cubical vessel is filled completely with 2M H_2SO_4 solution, whereas, spherical and cylindrical vessels are empty. (Take $\pi = 3$, Atomic mass of Br = 80)

17. Total content of cubical vessel is placed in cylindrical vessel and it is further filled completely with 1M BaCl_2 solution. Molarity of SO_4^{2-} ions in aqueous solution will be (BaSO_4 is completely insoluble in water)-
- (A) 1M (B) 0M (C) 0.66M (D) 0.33M
18. Now total content of the cylindrical vessel is placed in the spherical container (after filtration of BaSO_4 precipitate) & it is further filled completely with 32.4 % w/v of HBr solution molarity of H^+ in final solution will be-

- (A) $\frac{7}{4}$ M (B) 2M (C) $\frac{3}{2}$ M (D) $\frac{5}{2}$ M

Space for Rough Work

PART-3 : MATHEMATICS**SECTION-I(i) : (Maximum Marks : 18)**

- This section contains **FIVE** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct.
- For each question, darken the bubble corresponding to the correct option in the ORS.
- For each question, marks will be awarded in one of the following categories :
Full Marks : +3 If only the bubble corresponding to the correct option is darkened.
Zero Marks : 0 If none of the bubbles is darkened.
Negative Marks : -1 In all other cases

1. Sum of all the integers that do not belong to range of $y = \frac{11x^2 + 12x + 6}{x^2 + 4x + 2}$, is-

- (A) -7 (B) -9 (C) -4 (D) -2

2. The general solution of the equation $\sin^2 2x - \cos^2 8x = \frac{1}{2} \cos 10x$, is

(A) $\left\{ (2n+1)\frac{\pi}{20} \right\} \cup \left\{ \frac{n\pi}{3} \pm \frac{\pi}{9} \right\}, n \in \mathbb{I}$ (B) $\frac{n\pi}{3} \pm \frac{\pi}{11}, n \in \mathbb{I}$

(C) $(2n+1)\frac{\pi}{12}, n \in \mathbb{I}$ (D) $\frac{n\pi}{5} \pm \frac{\pi}{15}, n \in \mathbb{I}$

Space for Rough Work

3. In a sequence $\{a_n\}$, $a_1 = 2$, $a_{n+1} = 1 - \frac{1}{a_n}$ for $n \geq 1$. Let P_n be the product of its first n terms, then the value of P_{2020} is
- (A) $\frac{-1}{2}$ (B) -2 (C) $\frac{1}{2}$ (D) 1
4. Let P be a point on $y = x$ and Q be point on $x^2 - y^2 = 1$ and slope of line PQ is -1 , then the locus of mid points of PQ is -
- (A) circle (B) parabola (C) ellipse (D) hyperbola
5. The locus of feet of perpendicular from either foci of the ellipse $(x - y + 1)^2 + (2x + 2y - 6)^2 = 20$ on any tangent will be :
- (A) $x^2 + y^2 + 2x + 4y + 5 = 0$ (B) $x^2 + y^2 + 2x + 4y - 5 = 0$
- (C) $x^2 + y^2 - 2x - 4y - 5 = 0$ (D) $x^2 + y^2 - 2x - 4y + 5 = 0$
6. $y = \frac{2x+3}{x+1}$ is a rectangular hyperbola. Length of its latus rectum is -
- (A) 1 (B) $\sqrt{2}$ (C) 2 (D) $2\sqrt{2}$

Space for Rough Work

SECTION-I(ii) : (Maximum Marks: 32)

- This section contains **EIGHT** questions.
- Each question has **FOUR** options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
- For each question, choose the correct option(s) to answer the question.
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.
Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct options.
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks : -2 In all other cases.
- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

7. Let $\log_2(2^{x-1} + 3^{x+1}) = 2x - \log_2 3^x$. Then which of the following is/are true

(A) $x = \frac{1}{(1 - \log_2 3)}$ (B) $x = \frac{1}{(1 - \log_3 2)}$ (C) $x = \frac{\log_3 2}{\log_3(2/3)}$ (D) $x = \frac{\log_2 3}{\log_2(3/2)}$

8. Parallel tangents are drawn to the ellipse $x^2 + 2y^2 = 2$ such that distance between them is $\sqrt{5}$.

If quadrilateral ABCD formed by these tangents, then select correct option(s) -

(A) ABCD is a rhombus

(B) ABCD is a parallelogram

(C) Area of quadrilateral ABCD is $\frac{10}{\sqrt{3}}$

(D) Area of quadrilateral ABCD is $\frac{20}{\sqrt{3}}$

Space for Rough Work

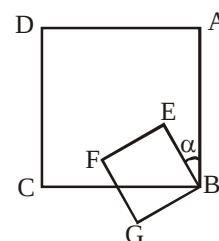
9. Let 'O' be the origin and the line $y = (2 - \sqrt{3})x$ cuts the line $x = 1$ at point A. Right isosceles triangle OAB is constructed such that hypotenuse is OA and B lies in first quadrant. Which of the following statements is (are) true ?
- (A) length $AB = \sqrt{3} - 1$ (B) length $OB = \sqrt{3} + 1$
- (C) $B \equiv \left(\frac{\sqrt{3}-1}{2}, \frac{3-\sqrt{3}}{2} \right)$ (D) $B \equiv \left(\frac{\sqrt{3}+1}{2}, \frac{3+\sqrt{3}}{2} \right)$
10. If the equation of circle S is given by $x^2 + y^2 + 4(\lambda + 1)x - y(8 + \lambda) + 6 - 6\lambda = 0$ then which of the following option(s) is/are true.
- (A) Centre of circle $S = 0$, always lie on a line $x + 4y = 18$
- (B) Centre of circle $S = 0$, always lie on a line $x + 4y = 14$
- (C) If $S = 0$ is orthogonal to $x^2 + y^2 + 2x + 4y + 1 = 0$ then $\lambda = \frac{19}{8}$
- (D) If $S = 0$ is orthogonal to $x^2 + y^2 + 2x + 4y + 1 = 0$ then $\lambda \in \phi$
11. If the system of equations
- $$2x - 3y + 5z = 12$$
- $$3x + y + \lambda z = \mu$$
- $$x - 7y + 8z = 17$$
- has infinite solutions then $2(\lambda + \mu)$, is divisible by
- (A) 3 (B) 6 (C) 11 (D) 7

Space for Rough Work

12. In a ΔABC , with usual notation r_1, r_2, r_3 are in harmonic progression. If its area is 24 and perimeter is also 24 then length of sides can be
 (A) 6 (B) 8 (C) 10 (D) 12
13. Consider $A_1, A_2, A_3, \dots, A_n$ and $B_1, B_2, B_3, \dots, B_n$, $n \geq 20$ are two different arithmetic progression

such that $\frac{A_n}{B_1} = \frac{\sum_{i=1}^n 2A_i}{\sum_{i=1}^n B_i} = \frac{B_n}{A_1} = 4$, then select the correct options

- (A) $\frac{B_{10} - B_8}{A_{12} - A_{11}} = \frac{1}{13}$ (B) $\frac{A_n}{B_n} = \frac{7}{2}$ (C) $\frac{A_n}{B_n} = \frac{53}{104}$ (D) $\frac{B_{10} - B_8}{A_{12} - A_{11}} = \frac{2}{13}$
14. ABCD is a square. BEFG is another square drawn as shown in figure with common vertex B such that E & F fall inside square ABCD. Then $\frac{DF}{AE}$ is equal to -



- (A) 1, when $\alpha = 30^\circ$ (B) $\sqrt{2}$, when $\alpha = 15^\circ$
 (C) $\sqrt{3}$ (D) 2

Space for Rough Work

SECTION-I(iii) : (Maximum Marks : 12)

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- For each question, darken the bubble corresponding to the correct option in the ORS.
- For each question, marks will be awarded in one of the following categories :
Full Marks : +3 If only the bubble corresponding to the correct answer is darkened.
Zero Marks : 0 In all other cases.

Paragraph for Questions 15 and 16

Let $\alpha, \beta, \gamma > 0$, $\alpha + \beta + \gamma = \frac{\pi}{2}$, $a = 5 + \tan\alpha \tan\beta$, $b = 5 + \tan\beta \tan\gamma$, $c = 5 + \tan\gamma \tan\alpha$.

15. The value of $(\Sigma \cot 2\alpha \cot 2\beta)(\Sigma a)$ is-

- (A) 15 (B) 16 (C) 1 (D) 0

16. Maximum value of $\sqrt{a} + \sqrt{b} + \sqrt{c}$ is-

- (A) $4\sqrt{3}$ (B) $3\sqrt{5}$ (C) 5 (D) $\frac{4}{\sqrt{3}}$

Space for Rough Work

Paragraph for Questions 17 and 18

If the locus of the circumcentre of a variable triangle having sides y-axis, $y = 2$ and $\ell x + my = 1$, where (ℓ, m) lies on the parabola $y^2 = 4x$, is curve C, then

On the basis of above information, answer the following questions :

17. Vertex of curve C is -

- (A) $\left(-2, \frac{3}{2}\right)$ (B) $\left(-2, -\frac{3}{2}\right)$ (C) $\left(2, \frac{3}{2}\right)$ (D) $\left(2, -\frac{3}{2}\right)$

18. Length of smallest focal chord of the curve C is -

- (A) $\frac{1}{4}$ (B) $\frac{1}{12}$ (C) $\frac{1}{8}$ (D) $\frac{1}{16}$

Space for Rough Work

Space for Rough Work

QUESTION PAPER FORMAT AND MARKING SCHEME :

16. The question paper has three parts : Physics, Chemistry and Mathematics.

17. Each part has one section as detailed in the following table.

Section	Que. Type	No. of Que.	Category-wise Marks for Each Question				Maximum Marks of the section
			Full Marks	Partial Marks	Zero Marks	Negative Marks	
I(i)	Single correct option	6	+3 If only the bubble corresponding to the correct option is darkened	—	0 If none of the bubbles is darkened	–1 In all other cases	18
I(ii)	One or more correct option(s)	8	+4 If only the bubble(s) corresponding to all the correct option(s) is(are) darkened	+1 For darkening a bubble corresponding to each correct option, provided NO incorrect option darkened	0 If none of the bubbles is darkened	–2 In all other cases	32
I(iii)	Paragraph Based (Single correct option)	4	+3 If only the bubble corresponding to the correct option is darkened	—	0 In all other cases	—	12

NAME OF THE CANDIDATE

FORM NO.

I have read all the instructions and shall abide by them.

Signature of the Candidate

I have verified the identity, name and Form number of the candidate, and that question paper and ORS codes are the same.

Signature of the Invigilator